PROGRAMMABLE WEARABLES FOR HUMAN PERCEPTION

MULTIMODAL ACQUISITION FOR ROAD SAFETY

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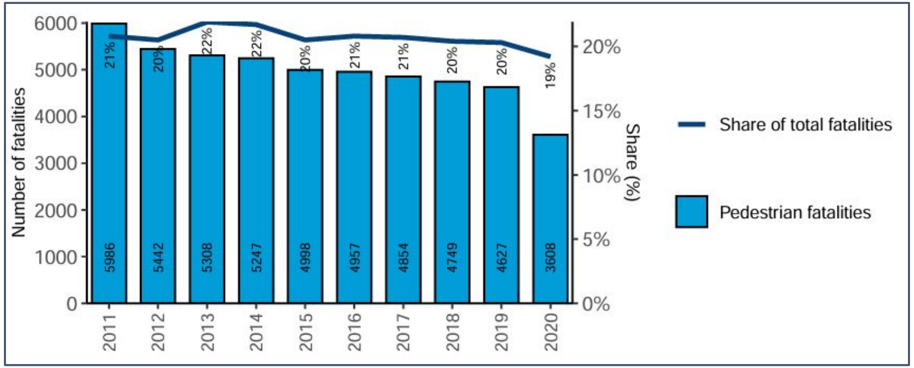
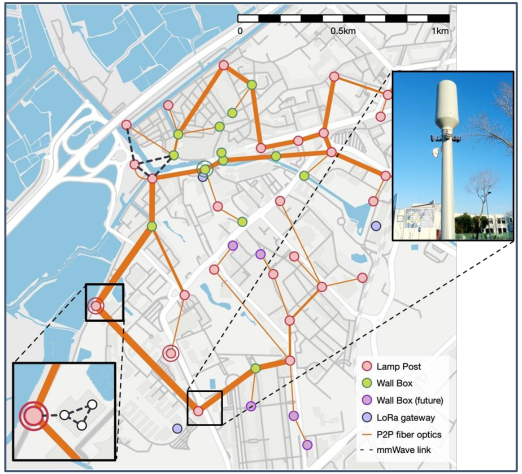
MOTIVATION

Protecting VRUS in Urban Environments

- **VRUs are the main target**
  - Pedestrians remain **highly exposed** in urban traffic scenarios.
- **Infrastructure alone is not enough**
  - **Limited** coverage
  - **Blind** spots
  - High deployment **cost**
- **AR-based solutions validated the concept**
  - **Bulky** AR solutions are not practical for **daily** use
- **Wearables open a more practical path**
  - **Discreet** sensing
  - **Continuous** monitoring
  - **Real-time** alerts





USE CASES

Crossing Intention

- Wrist IMU, GPS , and glasses **camera** infer **crossing intention**, alerting vehicles via **V2X**.

IMU WRIST/ WRISTBAND

_VELOCITY 0 KM/H
_RELAXED ARM

DIRECTION OF GAZE

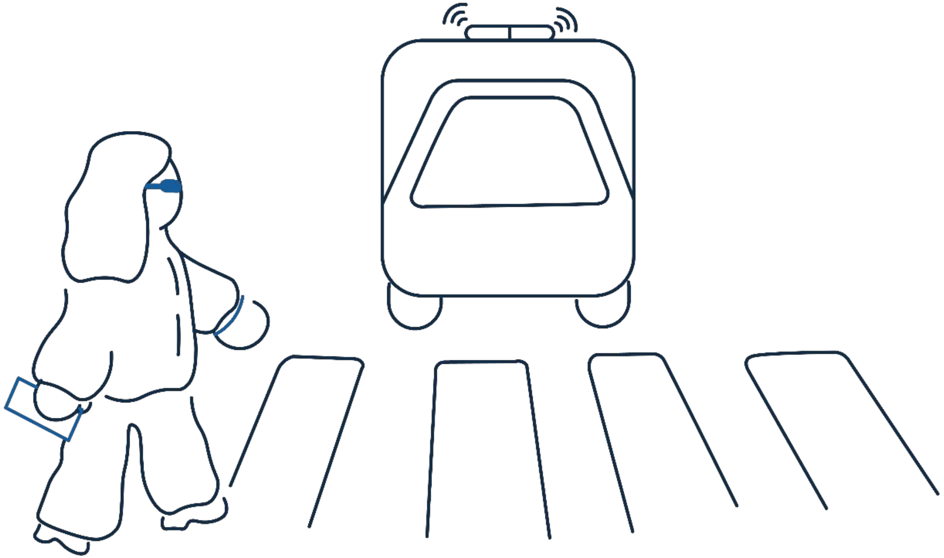
_CARRIAGEWAY

DISTANCE

_1.2 M OF CROSSWALK

INTENTION OF CROSS DETECTED

_CONFIDENCE 87%



USE CASES

Distracted Pedestrian

- **Microphone** detects a siren + **IMUs** confirm distraction + **GPS/accelerometer** confirm movement → **vibration** on the wrist.

MICROPHONE (GLASSES)

_SIREN DETECTED
_MABULANCE APPROACHING

IMU HEAD (GLASSES)

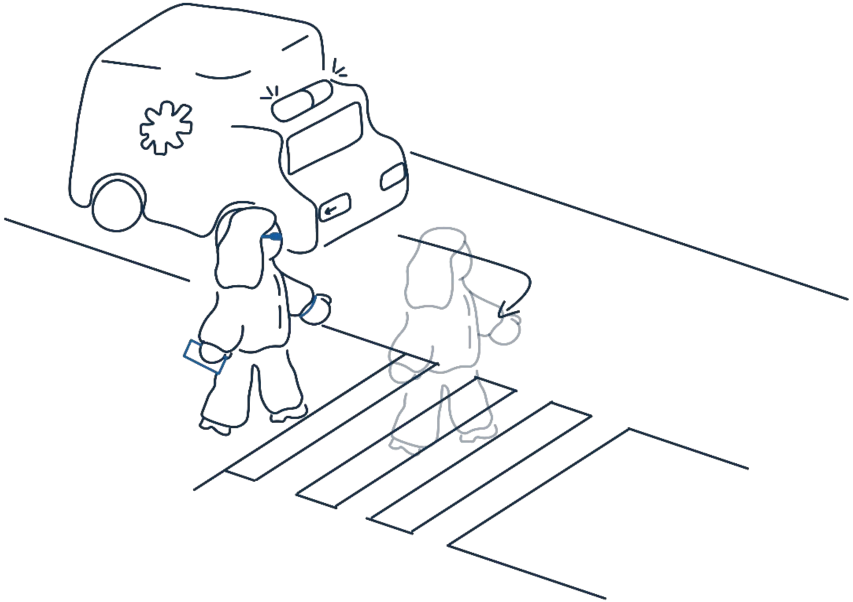
_NO LATERAL ROTATION DETECTED
_THE GAZE DINDN´T CHECK THE SIDES

IMU WRIST/ WRISTBAND

_CONTINUOS MOTION
_4.8 KM/H
_NO SLOW DOWN

SMARTPHONE ACCELEROMETER

_ONGOING MOVEMENT
_WITHOUT STOPPING

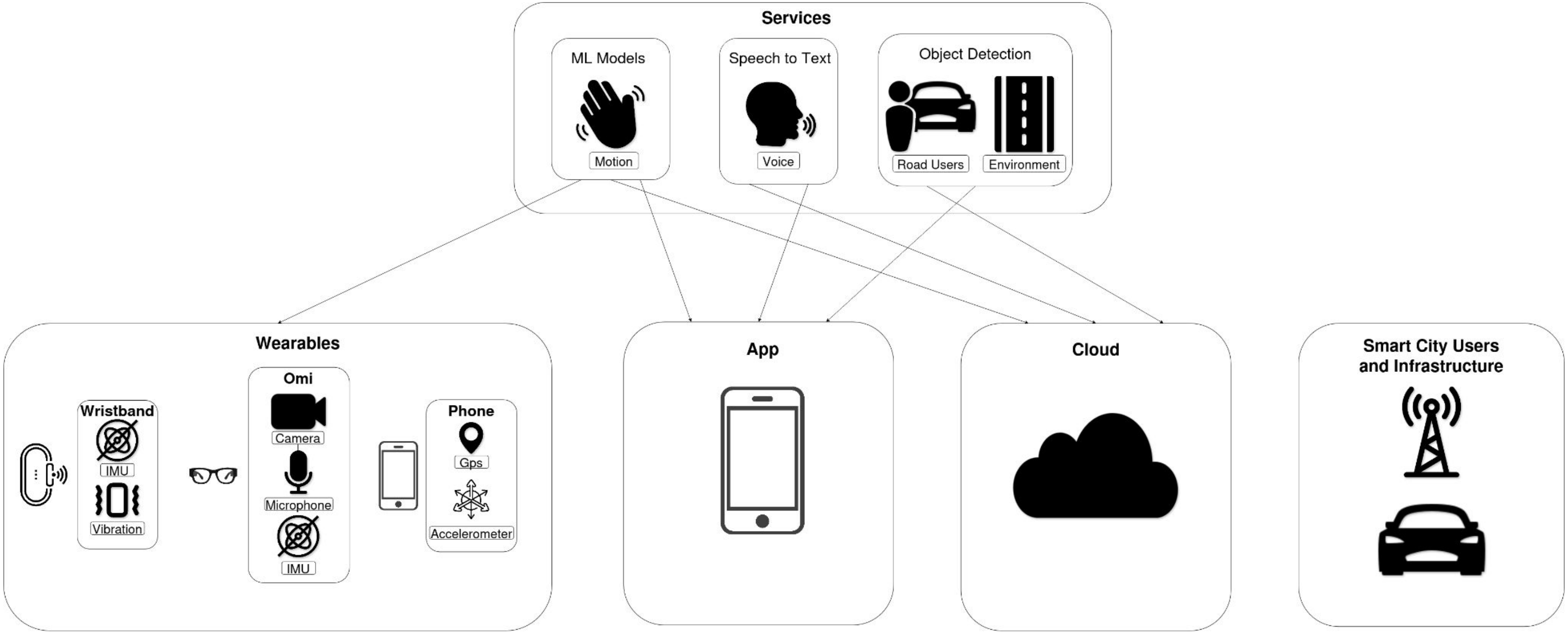


TIMELINE UI

_ SIREN DETECTED BY MICROPHONE → _PEDESTRIAN NEITHER SLOWS DOWN NOR LOOKS → _IMMEDIATE HAPTIC ALERT ON THE WRIST

ARCHITECTURE

System Architecture



Video Camera

What has already been achieved

- We acquire **camera** frames to get road **context** through **YOLO**
- Camera data **streaming** and photo capture working via **BLE** and **Wi-Fi**
- **Multi-configuration** YOLO
  - App **YOLOv8n**
  - Cloud **YOLO11n**
- Using **Brilliant Wear's** closed-source firmware to test different camera **configurations**
  - **Resolution**
  - Quality of the **jpg**
  - Packet **rate**

YOLO OFF: 480x360, Quality 74, Rate 10ms, ~7 FPS

YOLO ON: 480x360, Quality 74, Rate 10ms, ~4 FPS

IMPLEMENTATION

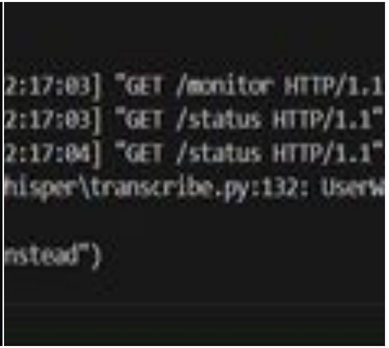
Microphone

What has already been achieved

- Two-layer approach
  - **Command**-oriented audio for **interaction**
  - **Context**-oriented audio for **situational** awareness
- Microphone data **acquisition** already working in the app via **BLE** and **Wi-Fi**
- Whisper already **integrated** into the audio **pipeline**
  - Cloud, **Whisper base, ~120-200ms**

Future Work

- Evaluate **different** microphone **configurations** and Whisper **model** setups
- Capture audio only when **relevant** sound is **detected**



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IMPLEMENTATION

IMU

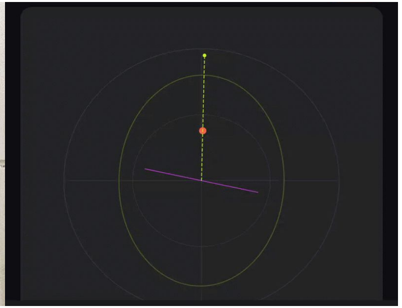
IMU

- Glasses

- **Real-time** streaming of **motion** sensor data
- **Head tracking** integration in the Android app

- Wristband

- The system base was **implemented** to support execution in **different** locations
- Different **ML models** were created for **device**, **app**, and **cloud**



Future Work

- Glasses

- **Improve** heading and trajectory **stability**
- **Strengthen** IMU + GPS fusion

- Wristband

- Compare total **response time** between **device**, **app**, and **cloud**
- Analyse the **impact** of each approach on system **performance**

Classes (Top scores)

```
Top class: idle (81.3%)

idle: 81.25%
walk: 18.75%
```

IMPLEMENTATION

Depth Anything

Implemented

- Depht to anything as a **tool** to obtain **distance** from objects or **depth** data
- **Depth** and **object detection** combined in a single functional flow
- **2 visualization** modes overlaid on the camera preview: **relative depth** colormap and **metric distance** in metres

Future Work

- **Improve** distance estimation **accuracy**
- Planned **migration** to run Depth Anything in the **cloud**, offloading computation from the **device**

640x640, ~2FPS

EVALUATION

Latency Evaluation Setup

Test Setup

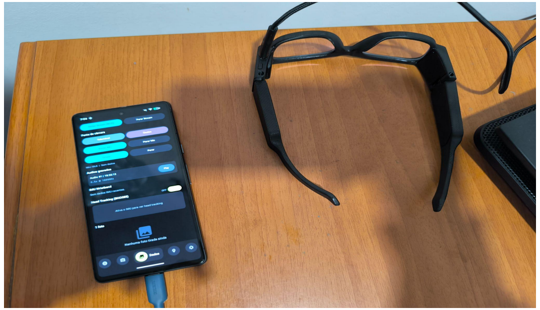
- Omi **glasses** facing a **fixed** target
- Constant **setup** maintained across **all** tests
- **Continuous** power **supplied** to the glasses
- **BLE** and **Wi-Fi** evaluated under the **same** conditions

Latency Measurement

- **End-to-end** latency measured from **capture request** to frame **reception** in the **app**
- Internal **timestamps** used for **each** frame
- **Fallback** timing applied when capture **status** was **unavailable**

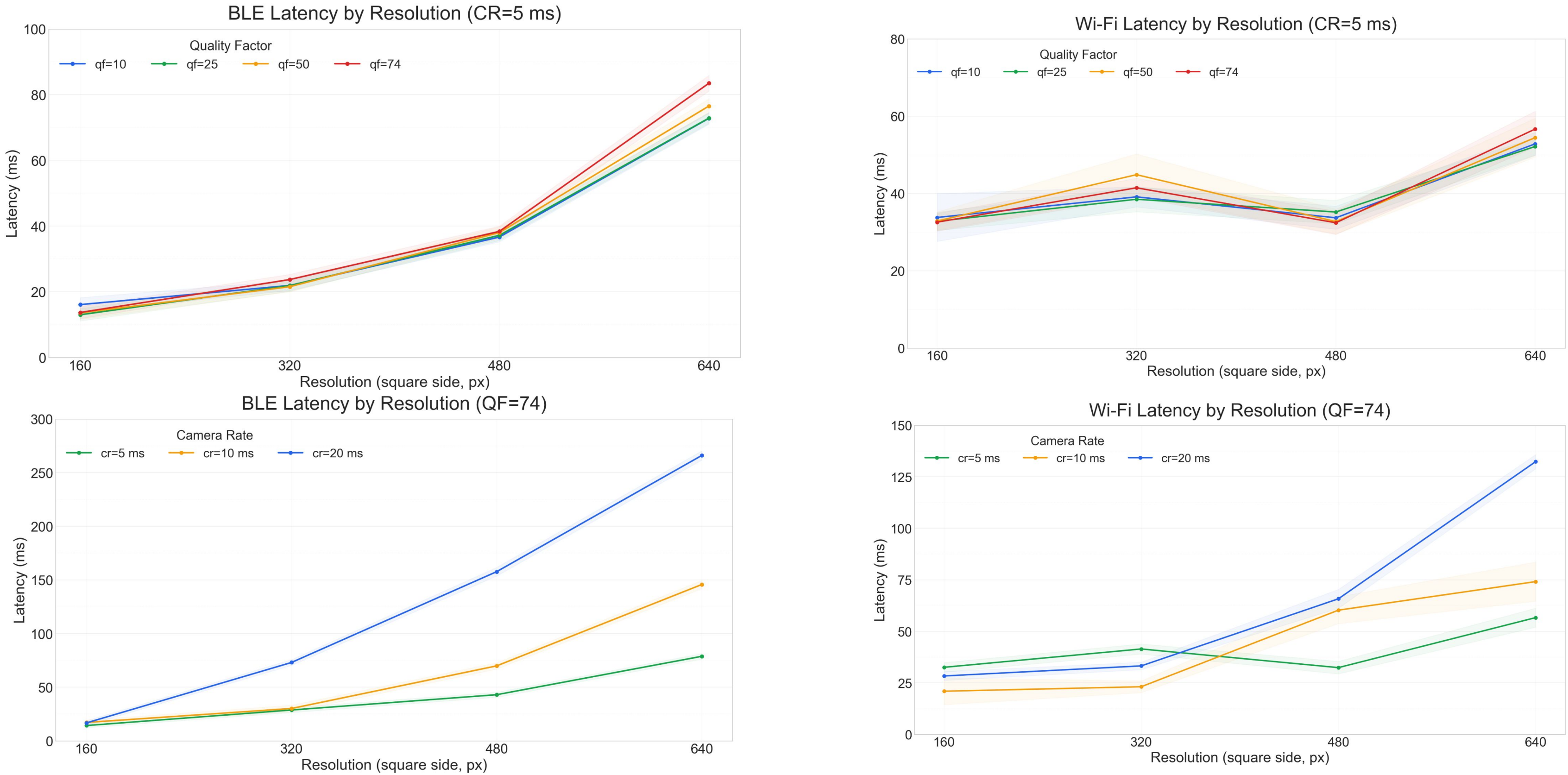
Scenario Parameters

- **Resolution**
- **Quality factor**
- **Camera rate**
- **1500 frames per scenario**



EVALUATION

Results



FIRMWARE

In-House Firmware

Implemented changes

- Closed source vs open source
- Local **Wi-Fi** communication **available** on the glasses
- **UDP** data transmission **available**
- **Simultaneous** image **capture** and **transmission**
- **Camera** and **IMU** communicate with the **ESP** over two independent **I2C** channels

Trade offs

- **Brilliant Wear** is plug and play, but our firmware, though less straightforward achieves better latency
- **Coupled** vs **decoupled. Adapting** the architecture to our **needs**

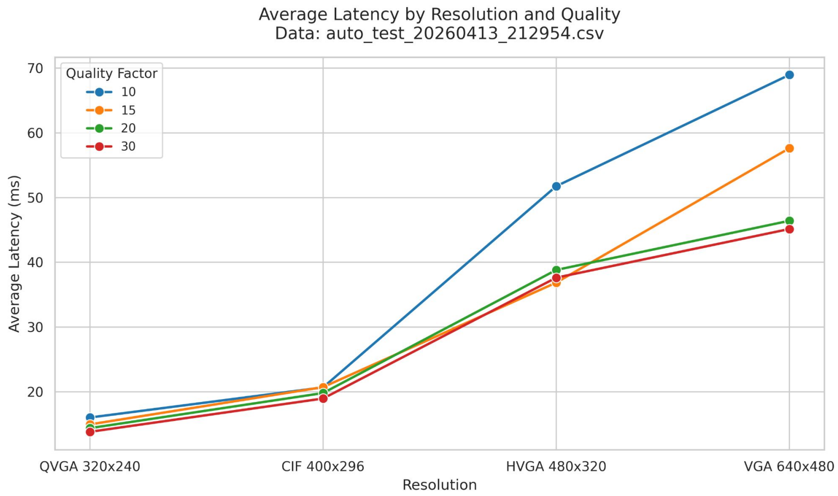
Evaluation and Next Steps

Next Steps

- Achieve the basic functionalities to test **use cases** on both **firmwares**
- Test **latency** in each processing **location**
- Compare total **response time** between **device**, **app**, and **cloud**
- Analyse the **impact** of each approach on system **performance**

Evaluation

- Overall **Lower latency** results compared to **Brilliant Wear** Firmware
- Supports **higher image quality** while having a lower latency



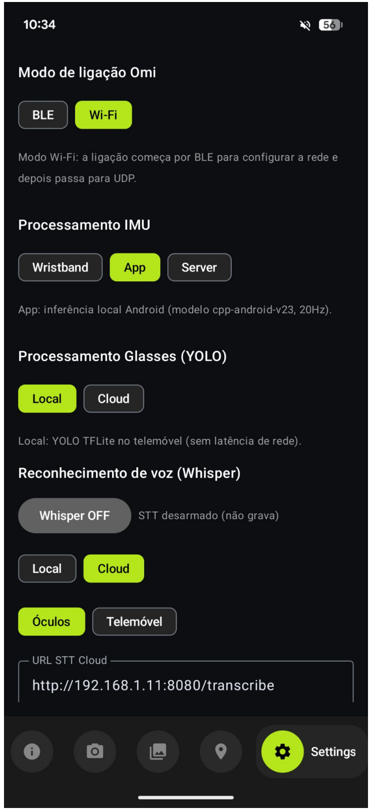
BWear:~7 FPS

Ours:~25 FPS

CONCLUSION

Conclusion and Future Work

- **Unified foundation achieved**
  - **Unified multimodal** sensing framework for low-powered **wearables**
  - Common **pipeline** for **acquisition**, **communication**, and **processing**
- **End-to-end evaluation still needed**
  - **Latency** evaluation across communication **settings** and processing **locations**
  - **Benchmarking** of remaining **configurations**, including **YOLO** and **Whisper** pipelines
- **Multimodal data fusion is the next key step**
  - Combine **video**, **audio**, **IMU**, and context data into higher-level **events**
  - Improve **robustness** and **reliability** of detections
- **Final goal: apply everything to the target use cases**
  - **Crossing intention detection**
  - **Distracted pedestrian alert**



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Thank You for Your Attention

Questions and Answers (Q&A)